Mehrab Beikzadeh

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Summary

I am a fifth-year Ph.D. candidate in Computer Science at the University of California, Los Angeles. My research intersects machine learning, health informatics, natural language processing, and Generative AI with a focused expertise in multimodal applications that combine text, image, and sensor data for comprehensive analysis and inference. I am an expert in designing and training deep learning models, such as CNNs, RNNs, and Transformers, and employing Transfer Learning techniques (LoRA, RAG, etc) with foundation models like LLMs, VLLM, as well as Reinforcement Learning with Human Feedback (RLHF), and in traditional machine learning algorithms.

Work Experience (Internship)

Google — Mountain View, CA

PhD Intern – Software engineering (Incoming Summer)

AstraZeneca — Gaithersburg, MD

PhD Intern - Data Science (GenAI Knowledge Management) (06/16/2025 – 09/19/2025)

• Built a graph-RAG pipeline on biomedical documents, using prompt-based entity masking, co-occurrence weighting, and community detection to generate Cypher queries that dynamically retrieve relevant attributes for LLM-driven chatbot question answering

Merck & Co — Cambridge, MA - Boston, MA

PhD Intern - Natural Language Processing (NLP) Research (06/17/2024 – 09/20/2024)

• Designed an improved relation extraction model (BERT, LLMs) for biomedical entities from PubMed abstracts, replacing the company's production model.

Education

University of California, Los Angeles (UCLA), Los Angeles - (09/202021 – Present)

Ph.D. in Computer Science

University of California, Los Angeles (UCLA), Los Angeles – (09/20/2021 – 12/15/2023)

M.S. in Computer Science, GPA: 3.8/4.00

Sharif University of Technology (SUT), Tehran – (09/22/2016 – 09/31/2020)

B.S. in Electrical Engineering, GPA: 3.97/4.00, Ranked 3rd in the department.

Ranked 3rd in KONKUR: National University Entrance Exam among 500 thousand competitors, (07/14/2016)

Research Experiences

Reinforcement Learning for Clinical LLMs in Mental Health (Oct. 2024 - Present)

- Enhancing LLMs for psychotherapy using reinforcement learning with clinical and patient feedback.
- Building a clinical preference data repository and an RLHF toolkit for scalable model adaptation.
- Contributing to evaluations of LLM-based interventions for mental health.
- Developing a scalable multi-objective DPO framework for aligning LLMs across multiple dimensions.

Using NLP for Relation Extraction in Drug Discovery (June. 2024 – Sept. 2024)

- Led the development of deep learning models, including PubMedBERT and BlueBERT, for Named Entity Recognition (NER) and Relation Extraction (RE) from PubMed abstracts.
- Applied GPT-4 with few-shot learning, structured in JSON, for optimal performance and system integration.

Analysis of Risky Behaviors via Social Media Activity in MSM Populations (Apr. 2021 - Present)

- I led the development of an Android application using Ktolin and Java, designed to collect crucial text and location data, with backend services on AWS.
- I apply advanced Machine Learning and NLP techniques, including traditional NLP techniques as well as fine-tuning Language Models (LLMs) to analyze textual data and extract insights on risky behaviors, using Python for model training and analysis.

Analysis of Homophobic Sentiments & Monkeypox Discussions on Twitter (Apr. 2021 – Jul. 2023),

- I conduct extensive data analysis on large Twitter datasets, focusing on the interaction between words related to homophobic sentiments and monkeypox discussions, using SQL for data manipulation and Python for analysis.

Mental Health Analysis Using Smartwatch Technology (Oct. 2021 – Dec. 2022)

- I design and implement a Garmin watch app using Monkey-C, enabling real-time survey capture, and I develop a companion Android app for seamless data collection and synchronization.
- I used Firebase as the backend database, ensuring efficient and scalable data management for the collection and synchronization of real-time survey data.
- I employed contrastive learning as a self-supervised method to pre-train models on large unlabeled datasets for further fine-tuning. This innovative approach allowed for effective model initialization, significantly improving the performance on subsequent tasks with minimal labeled data.

Enhancing AI-driven Healthcare Communications via Privacy-Preserving Technologies (Mar. - May 2023)

- Developed a privacy solution using Langchain to ensure confidentiality in ChatGPT healthcare applications.

Selected Publications

[1] **Mehrab Beikzadeh**, et al. "Identifying Substance Use and High-Risk Sexual Behavior Among Sexual and Gender Minority Young People Using Mobile Phone Data: Development and Validation Study " *JMIR mHealth and uHealth* , 2025.

[2] **Mehrab Beikzadeh**, et al. "Multi-Objective Alignment of Language Models for Personalized Psychotherapy." arXiv preprint, 2026.

[3] Anaelia Ovalle, **Mehrab Beikzadeh**, et al. "ChatGPT for Us: Preserving Data Privacy in ChatGPT via Dialogue Text Ambiguation to Expand Mental Health Care Delivery" *arXiv preprint arXiv:2306.05552*, 2023.

[4] Brian TaeHyuk Keum, Chenglin Hong, **Mehrab Beikzadeh**, et al "Letter to the Editor: Mpox Stigma, Online Homophobia, and the Mental Health of Gay, Bisexual, and Other Men Who Have Sex with Men" *LGBT Health*, 2023.

Skills

Programming: Python, PyTorch, TensorFlow, Langchain, Matlab, C, C++, Java, Kotlin, Objective-C, Swift, SQL

Methods: NLP, Transformers, Foundation Models, LLM, VLLM, Deep Learning, RNN, RAG, QLoRA, Prompt Engineering, Bayesian Analysis, Contrastive Learning, Random Forest, SVM, Logistic Regression, Gradient Boosting, etc.

Circuit Design: FPGA, Verilog, PCB, PSPICE, HSPICE, Altium Designer, Proteos, Model-sim

Tools and Software: AWS**,** Android Studio, Eclipse, Simulink, Multisim, Firebase, Linux, Git, Docker, Flask, Django

Soft Skills: Self-motivated, Presenter, Team player, Communicative, Collaborative, Innovative, Problem-solver, Mentor

Honors and Awards

Ranked 3rd: Ranking in the Electrical Engineering Department among students graduating in 2020, (2020)

Ranked 1st: Ranking in the Electrical Engineering Department among about 200 students in the third semester, (2017)

Bronze Medal: National Mathematical Olympiad, (2015)

Semi-finalist: National Olympiad in Mathematical 3 times, in Physics and Computer Science 2 times, (2013-2015)

Ranked 3rd: Programming Competition among competitors from all universities in Iran which was held in SUT, (2016)

GRE Quantitative Reasoning: 170/170, (2019)

Selected Courses

Deep Generative Models (Audited) - Neural Network and Deep Learning(A) - Machine Learning Algorithms(A+) - Algorithms and Complexity(A) - Algorithmic Machine Learning(A+) -IOT Connectivity(A) - Software Engineering(B+)- Computer Network Fundamentals(B+)- Natural Language Processing (Audited) - Bayesian Networks(A) - Fundamentals of Programming: 20.0/20.0 - Numerical Calculation (19.8/20.0) - Probability and Statistics(19.6/20.0) - Communication Systems(19.5/20)

Experiences(Teaching Assistant):

Algorithms and Complexity (CS180) - Digital Design Laboratory (CS152a) (8 times) - Probability and Statistics